MTL1002: TUTORIAL 1
(LINEAR ALGEBRA)

(1) Consider the following system of three linear equations with real coefficients and in two unknowns:

$$\begin{aligned} a_1x + b_1y &= c_1, \\ a_2x + b_2y &= c_2, \\ a_3x + b_3y &= c_3. \end{aligned}$$

Interpret geometrically the following statements:
(a) The system has no solutions.
(b) The system has a unique solution.
(c) The system has infinitely many solutions.
Further, provide values to $a_i, b_i, c_i \in \mathbb{R}$ $(i = 1, 2, 3)$ so that the above statements hold.

(2) Consider the following system of three linear equations with real coefficients and in three unknowns:

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1, \\ a_2x + b_2y + c_2z &= d_2, \\ a_3x + b_3y + c_3z &= d_3. \end{aligned}$$

Interpret geometrically the following statements:
(a) The system has no solutions.
(b) The system has a unique solution.
(c) The system has infinitely many solutions.
Further, provide values to a_i, b_i, c_i, d_i $(i = 1, 2, 3)$ so that the above statements hold.

(3) Consider the following system of two linear equations in three unknowns:

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1, \\ a_2x + b_2y + c_2z &= d_2. \end{aligned}$$

Can this system have a unique solution? Interpret the following statements geometrically.
(a) The system has no solutions.
(b) The system has infinitely many solutions.
Further, provide values to a_i, b_i, c_i, d_i $(i = 1, 2)$ so that the above statements hold.

(4) Consider a system of linear equations in complex coefficients.
(a) Can we interpret the statements in Questions 1, 2 and 3 exactly in the same way?
(b) Solve the following system of equations (for finding $x, y \in \mathbb{C}$):

$$\begin{aligned} (1-i)x + (1+i)y &= 2 + 3i, \\ (1+i)x + (1-i)y &= 3 - i. \end{aligned}$$

(5) Suppose the lines L_1 and L_2 are defined by

$$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3} \quad \text{and} \quad \frac{x-2}{1}=\frac{y-3}{2}=\frac{z-1}{3}.$$

(a) Prove that $L_1 \cap L_2$ is empty.
(b) Write a system of four linear equations in three unknowns in standard form which has no solutions using the fact that $L_1 \cap L_2$ is empty.

(6) Suppose A, B are square matrices of same size. Prove the following statements:
(a) $\operatorname{tr}(A+B)=\operatorname{tr}(A)+\operatorname{tr}(B)$ and $\operatorname{tr}(AB)=\operatorname{tr}(BA)$.
(b) $\det(AB)=\det(A)\det(B)$ (assume A and B are 2×2 matrices) and in particular, $\det(AB)=\det(BA)$.
(c) $\det(A+B)=\det(A)+\det(B)$ is false.

(7) Consider the following matrices:

$$\begin{pmatrix}1&0&5\\0&2&3\\0&0&0\end{pmatrix},\ \begin{pmatrix}1&0&5\\0&1&3\\0&0&1\end{pmatrix},\ \begin{pmatrix}0&1&2\\1&0&4\\0&0&0\end{pmatrix},\ \begin{pmatrix}0&1&0\\0&0&0\\0&0&1\end{pmatrix},\ \begin{pmatrix}0&1&0\\0&0&1\\0&0&0\end{pmatrix}.$$

(a) Which of the above matrices are in row reduced echelon form? Give a reason when the matrix is not row reduced echelon.
(b) Find row reduced echelon matrices row equivalent to each of the above matrices.

(8) Prove that every elementary matrix is invertible and the inverse is also an elementary matrix.

(9) Consider the following matrices:

$$\begin{pmatrix}1&1&2\\2&3&8\\-3&-1&2\end{pmatrix},\ \begin{pmatrix}1&2&-4\\-1&-1&5\\2&7&-3\end{pmatrix},\ \begin{pmatrix}1&3&-4\\1&5&-1\\3&13&-6\end{pmatrix}.$$

(a) Compute the rank of the above matrices.
(b) Determine which of the above matrices are invertible.
(c) For the invertible matrices, find their inverses by reducing the matrix to row reduced echelon form (identity matrix).

(10) Write the following matrices as the product of elementary matrices (if possible):

$$\begin{pmatrix}1&-3\\-2&4\end{pmatrix},\ \begin{pmatrix}1&2&3\\0&1&4\\0&0&1\end{pmatrix},\ \begin{pmatrix}1&1&2\\2&3&8\\-3&-1&2\end{pmatrix}.$$

(11) Solve the following systems of homogeneous linear equations by reducing the coefficient matrix into row reduced echelon form:
(a) $2x_1+4x_2-5x_3+3x_4=0,\ 3x_1+6x_2-7x_3+4x_4=0,\ 5x_1+10x_2-11x_3+6x_4=0.$
(b) $x-2y-3z=0,\ 2x+y+3z=0,\ 3x-4y-2z=0.$
(c) $x_1+2x_2+3x_3-2x_4+4x_5=0,\ 2x_1+4x_2+8x_3+x_4+9x_5=0,\ 3x_1+6x_2+13x_3+4x_4+14x_5=0.$

(12) Solve the following systems of equations by reducing the augmented matrix to the row reduced echelon form:

(a) $x_1 - x_2 + 2x_3 = 1, \quad 2x_1 + 2x_3 = 1, \quad x_1 - 3x_2 + 4x_3 = 2.$

(b) $x_1 + 7x_2 + x_3 = 4, \quad x_1 - 2x_2 + x_3 = 0, \quad -4x_1 + 5x_2 + 9x_3 = -9.$

(c) $x_2 + 5x_3 = -4, \quad x_1 + 4x_2 + 3x_3 = -2, \quad 2x_1 + 7x_2 + x_3 = -1.$

(d) $-2x_1 - 3x_2 + 4x_3 = 5, \quad x_2 - x_3 = 4, \quad x_1 + 3x_2 - x_3 = 2.$

(13) For real numbers a and b, consider the following system of equations:

$$x + 2y + z = 3, \quad ay + 5z = 10, \quad 2x + 7y + az = b.$$

(a) Find all values of a for which the system has a unique solution.

(b) Find all pairs (a, b) for which the system has more than one solution.

(14) Find $a, b, c, p, q \in \mathbb{R}$ such that the following system has a solution:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & p \\ 0 & 0 & q \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$